

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<thought >

To solve this problem , I need to recall the anatomy of the hand and locate the anatomical snuff box . The anatomical snuff box is a hollow between four tend ons , and it is a region of prominence on the palm . I need to find the main artery that passes through this region , as the bleeding could be from damage to the artery .

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< answer >

B . Ulnar artery </answer >

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The anatomical snuff box contains tend ons and nerves but does not pass through the arteries . The region is not supplied by the radial artery , nor by the brachial artery . The Prince ps pollicis artery , also not in the snuff box , serves a specialized area of the thumb rather than the hand in general . On the other hand , the ulnar artery passes through the anatomical snuff box , providing blood supply to the hand . Damage to this artery could explain the severe bleeding from the right hand . </thought >

